13 Assimilating Minds

State estimation is intimately connected with the meaning of life.
—Dan Simon, 2006 [Sim06a]

13.1 We Are All State Estimation Machines

It is a quiet secret of most neuroscientists and neurophysicians that they were drawn to the study of the brain in the hope of understanding the mind. Understand the mind and you understand life.

Coming to grips with the reality that, in our lifetimes at least, we will never understand the mind was a blow to most of us. The profusion of information about the nervous system has become overwhelming in an information age, where even high-quality publications about the brain and mind expand beyond the reach of any one of us to keep track of.

So it comes with some initial surprise to find, at the end of one of the best texts on state estimation yet written, a chapter on the relationship between state estimation and the meaning of life [Sim06a].

Similarly, in reading deeply within the geophysical literature, one stumbles on a quite serious discussion: "Like a data assimilation system, the human mind forms a model of reality that functions well, despite limited sensory input, and one would like to impart such an ability to the computational model" [DTW06].

The opportunity to speculate on these broader issues, prior to confirmed senility, is an opportunity that the author cannot pass on. I hope that these thoughts are welcomed by any reader who has had the patience and long-suffering temperament to have come this far in this book.

13.2 Of Mind and Matter

[The mind] can never produce any new quality in the object, which can be the model of that idea, yet the observation of this resemblance produces a new impression in the mind, which is its real model.
—David Hume, 1748 [Hum07]

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You walk outside and need to decide whether your perceptions are delivering to your mind sensations of a banana or an airplane.[1] The empirical Scottish philosopher David Hume, in his 1748 book, *An Enquiry Concerning Human Understanding* [Hum07], eloquently considered this problem (albeit without the airplanes):

> [N]othing is ever really present with the mind but its perceptions or impressions and ideas, and that external objects become known to us only by those perceptions they occasion. To hate, to love, to think, to feel, to see; all this is nothing but to perceive. [Hum07]

The mind is a state estimation machine, with regard to external objects as Hume speculated on, and internally, with regard to introspection. Just as in control engineering, there are many ways to accomplish state estimation. Kalman's original filter was optimal for linear systems with Gaussian noise. There is no rationale to suspect that our minds use Kalman's filter in its linear or linearized guise. There is no indication that evolution might have optimized our mental "filter" for the external world. We seem to use and demand a luxurious amount of energy to keep our minds running (about 20% of our basal metabolic rate) [Len03].

But filters are essential. As discussed this book, dynamical data assimilation filters are really filters. They remove noise in an effort to extract the relevant dynamics they seek. Simon speculates on the moral philosophy and theology of such filters [Sim06a]. In the theory of the cognitive dynamics of schizophrenia, the inability to parse and filter the incoming sensory stream has been postulated to be at the core of the symptoms, and correlative physiological data support this view [NAW+91]. Optical illusions [HP05] might be viewed as reflective of our error-prone reconstructions of the external world. We think that normal minds have more effective (more optimized?) filters than disordered ones.

13.3 Robot Beliefs versus Cogito Ergo MRI

> We shall adopt an approach that we call model-dependent reality. It is based on the idea that our brains interpret the input from our sensory organs by making a model of the world.
> — Stephen Hawking and Leonard Mlodinow [HM10]

In control robotics, the robot has a model of its world—perhaps a map it needs to navigate to perform its tasks. It takes in sensor data in various forms: video, laser range finding, global positioning system, and so on. And then it forms an estimate of where it thinks it is—a *belief*, in control jargon [TBF05].

When our minds are intact, we also estimate where we are and navigate. When our minds are severely damaged, we may not give external indications that we are conscious.

1. This phrase was shamelessly lifted from Dan Simon's discussion [Sim06a].

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Nevertheless, in recent years, we have developed the unprecedented ability to peer within people's minds with functional magnetic resonance imaging (fMRI). Patients who survive severe traumatic brain injury typically emerge to a state of wakefulness within a period of weeks. This state may not give evidence of conscious response to internal thought or external interaction; it may be a state of wakefulness without apparent awareness of one's self or surroundings. Such states are described with the unfortunate term *vegetative*.

A recent report documented the case of a patient who was vegetative following a severe traumatic brain injury. On fMRI imaging, when asked to imagine navigating through all of the rooms of her house, she demonstrated activity in the parts of the brain involved in spatial imagery navigation [OCB$^+$06]. This finding has generated much discussion in our society. Is the act of mentally navigating (or a motor activity such as swinging a tennis racket) equivalent to consciousness? This evidence is perhaps not convincing, but too risky to dismiss in error.

More recently, Monti and colleagues [MVC$^+$10] described a patient initially diagnosed as vegetative[2] who was seen to display such navigational versus motor (tennis) activity on fMRI scanning. He was then taught to respond *yes* to one form of imagery and *no* to the other form of activity. He was able to demonstrate consistent yes-no responses to complex personal questions.

One must be careful in interpreting such measurements. We cannot make the "unjustified assumption that the association between a behavior and a pattern of brain activation implies the converse" [NH07]. Brain activation is a necessary but not sufficient condition for evidence of consciousness, unless you can show that the same activation cannot occur without it [NH07, Rop10].

I have been careful not to define consciousness in this discussion. But whatever that term might represent, whatever doubts we might have regarding navigation or swinging a racket, that such imagery might represent more complex thoughts raises difficulties in interpreting these phenonema to a higher level.

These findings bring us to one of the deep issues that artificial intelligence has grappled with. In 1950, Alan Turing [Tur50] suggested that perhaps a digital state machine—a digital computer—would be able to think by following the rules of digital computation. He discussed numerous ways of opposing or supporting this argument. John Searle's [Sea80] *Chinese room* argument is the most cited refutation of Turing's *test*. Program a computer to take written questions in a language—say Chinese. It replies in an apparently intelligent fashion just as a conscious person who spoke Chinese might. Then place someone in a room with the same computer program, and perhaps a library of reference books. The person, knowing nothing of the Chinese language, takes the questions and uses the

2. This patient, on reexamination could display "reproducible but highly fluctuating and inconsistent signs of awareness" at the bedside [MVC$^+$10]. Such a finding technically reclassifies the diagnosis from vegetative to the new term "minimally conscious state" [Far08].

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computer program to create answers just as a computer would. Although the responses appear intelligent, we know that the person knows absolutely nothing of Chinese. Is this intelligence?

We appear to be at the stage where people with severely damaged brains are now manipulating symbols much in the fashion that Turing and Searle were considering in their philosophical writings. Perhaps the issue is one of uncertainty and risk. We risk little if we are wrong about a machine's thoughts. But we simply cannot make a mistake in the face of the evidence of human thought and intention. I'll side with the humans.

13.4 Black versus Gray Swans

The works and customs of mankind do not seem to be very suitable material to which to apply scientific induction [Tur50]

Turing [Tur50] in his consideration of whether machines might think, nicely discusses the appalling mistakes we make by applying scientific induction to scientific problems. If you want to ask whether a machine can think, the reader might examine his or her past experience with machines. You may have seen tens of thousands of machines. They were ugly and stupid. Does induction, extrapolating into the future based on past experience, tell us anything useful? Often not.

A marvelous rediscovery and expansion of the fallacy of logical induction is discussed by Nassim Taleb [Tal07]. His poignant example of the turkey as logician is not to be missed. Although Taleb's exposition is focused on the tracking and prediction of economics, the lessons for our observation of brain dynamics are very pertinent indeed.

In the author's admittedly biased perspective, Taleb's black swans boil down to the issue of model inadequacy. if we do not have a fundamental model of a system that truly characterizes its dynamics, then we need to rely on statistical models based on previous empirical data. Empirical models, such as our discussion of principal orthogonal decomposition in chapter 7, can capture only what has been previously observed. The epitome of such empiricism was expressed by:

I will say that I cannot imagine any condition which could cause a ship to founder. I cannot conceive of any vital disaster happening to this vessel. Modern shipbuilding has gone beyond that.
—E. J. Smith, captain of the Titanic (New York Times, April 16, 1912)

Taleb is rightfully concerned with natural dynamics that have power-law distributions of their amplitudes. Such events can certainly have underlying generative models that are inherently unpredictable [Man82]. Having dynamics with extreme events need not necessarily require that the underlying events be unpredictable. But it does require that one has a grasp of those dynamics. Airframes and pendula can be effectively modeled. The short-term atmospheric dynamics of weather can be modeled. Parts of the brain appear to

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be modelable. Can the economy be better modeled? Taleb might be severely skeptical, but his induction here might not be better than our induction about black swans.[3]

There is actually a theory that all of physics is, in effect, a balance between measurement of the natural world and uncertainty in those measurements. We are playing an information-theoretic game with nature, seeking to solve an extremization problem, in which information we gain comes at the expense of nature's disorder, as the system is closed [FC96]. In Taleb's *extremistan*, the appearance of large errors invalidates our models. The stock market crash of 1929 and the credit market meltdown of 2008 certainly support Taleb's perspective on financial markets. But airframes rarely crash these days. Whether or not your system belongs to extremistan may depend on what the physics are. Perhaps the brain, or at least portions of the brain, are now tractable enough that the conjectures presented in this book are not doomed to fall prey to the biting skepticism of Taleb.

Taleb also points to a key problem for our work on state estimation here. What are the consequences of losing track of the system? Were errors and uncertainty described by the neat Gaussian error distributions we have discussed, as Kalman's original filter assumed or our unscented version presumed, then things would be simple. Nothing in life is Gaussian. Julier and Uhlmann [JU97a] dealt with this in part by introducing a fudge factor—their kappa—that boosted the distribution width of their uncertainty to take unknowns into account. Taleb would jeer at us: If we had an uncertainty distribution without a well-defined mean, and with long "tails," then wherever we set our Gaussian distribution we would be hit from the rear by low-flying black swans. Control theory has long recognized the danger of such fowl events, sparking a tremendous amount of energy into *robust control* strategies [ZDG96, Sim06a].

Perhaps the question comes down to the consequences of losing track of a system we are observing. If an airframe controller loses track of a landing sequence, the consequences are dreadful. If a Parkinson's disease controller loses track of the oscillations within the basal ganglia, the patient will get a bit stiffer, and we can use any number of strategies commonly employed in particle filtering [Sim06a] to spread out the ensemble of state estimates to recapture the trajectory. If a seizure controller loses track of a system, an epileptic seizure might result—serious, but likely not lethal in and of itself. I hear Taleb scoffing at us. To my knowledge, no one has attempted to take an ensemble Kalman filter, such as the unscented filter, and parameterized the uncertainty with a power-law distribution. There is nothing that would prevent such a maneuver were it indicated. It would certainly decrease the accuracy of our state estimations. But it would make our tracking in the face of non-Gaussian distributions in critical applications more robust. And "a gray swan concerns modelable extreme events" [Tal07].

3. There actually are black swans. But they are in Australia. Our literature and science had a few thousand years of certainty that all swans were white.

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The case for model-based observation and control rests on state estimation and short-term prediction. Taleb's cautions are for phenomena where our models are so inadequate, and the consequences so dire, that operating with such risk exposure is hazardous. Nevertheless, there is a good operational test to see whether or not you are in the insane danger zone: "Prediction, not narration, is the real test of our understanding of the world" [Tal07]. The author completely agrees, so long as the prediction horizon is short enough to encompass the nonlinearities of the system. Weather (as opposed to climate) prediction, beyond two weeks, remains pretty useless. But for the dynamical diseases we have discussed in this book, predictions by our models from tens of milliseconds to tens of seconds would be sufficient to accomplish much.

13.5 Mirror, Mirror, within My Mind

We may then lay it down for certain that every representation of a movement awakens in some degree the actual movement which is its object. . . .
—William James (*The Principles of Psychology*, vol. 2, 1890, p. 611 [Jam90])

In 1992, di Pellegrino and colleagues [dPFF$^+$92] were studying neurons in primates that fired in response to specific motor movements, and discovered that those same neurons fired in response to similar movements on the part of the investigators. They sparked a period of intense study of such effects, which have now been observed in an interlocking hierarchical set of regions in the brain related to vision and movement [ID06]. We now speculate that this system forms part of our normal ability to imitate someone else's motor skills, and perhaps more important, might critically underlie our social skills of empathy and understanding of others. In the normal state, we speculate that such mirror systems solve the *problem of other minds* [Iac09]. We have also speculated how cognitive disorders such as autism [OHM$^+$05, DDP$^+$06, RFDC09] and schizophrenia [AA05] might be disorders of such systems.

Mirror neurons appear to be state estimation systems. Karl Friston and colleagues have proposed that the mirror neuronal system functions as a predictive coding model [KFF07]. In this scheme, each portion of this hierarchical system, from lower-level motor output through higher-level goal generation, has feedback and error correction with a built-in computational model of each interacting system. Indeed, the interaction of the mind with its motor system—the only way it can communicate with the external world—and social interactions, are compelling [WDK03]. Similarly, in the visual system, such feedback of efferent copies of predicted events seem critical [DR05].

At each level in the motor or visual system, the type of state estimation described in this book appears to be important. Kalman's brain probably was not born with Kalman's filter already hard wired. But as the evidence of multiple levels of state estimation within the brain and between brains in society become stronger, nature appears to have given us some type of model-based filter.

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Friston's predictive coding is really an instantaneous prediction—a zero-time prediction horizon. Yet we live in a world where events and time move inexorably forward. Our motor interaction with the stream of events necessitates *prospective coding* [SBP07].

The juvenile sea squirt wanders through the sea searching for a suitable rock or hunk of coral to cling to and make its home for life. For this task, it has a rudimentary nervous system. When it finds a spot and takes root, it doesn't need its brain anymore, so it eats it! (It's rather like getting tenure.) The key to control is the ability to track or even anticipate the important features of the environment, so all brains are, in essence, anticipation machines. (Daniel Dennett, 1991 [Den91])

13.6 Carl Jung's Synchronicity

The psychologist Carl Jung deeply explored synchrony and the mind [JP55a]. Jung was inspired by an essay published in 1851 as part of a larger collection by Arthur Schopenhauer, *Parerga and Paralipomena* "Transcendent Speculation on the Apparent Deliberateness in the Fate of the Individual" [SP74]. Schopenhauer was a bit of an extreme determinist. He envisioned the events of the world as trajectories (meridians), whose intertwining at a particular instance of time might bring us simultaneity without causal connectedness:

Accordingly, all those causal chains, that move in the direction of time, now form a large, common, much-interwoven net which with its whole breadth likewise moves forward in the direction of time and constitutes the course of the world. [SP74]

Our estimates of our states within the stream of time and the events of the external world can be estimated only by what we have access to. Our efforts are abetted by our models of the inaccessible variables of the external world and future time through our internal models.

[L]ife in time can never properly be understood, just because no moment can acquire the complete stillness [*fuldelig Ro*][4] needed to orient oneself backwards. (Søren Kierkegaard, *Papirer, 1833–1843*, p. 441 [Kie69] Translation from [Han82])

Some of Schopenhauer's deterministic transcendent fatalism [SP74] is the same semantics that we face today with deterministic random number generators. All computer random number generators are deterministic (you would need to time radioactive decay, for instance, to see what we think is true randomness). But the numbers my computer spits out are, for all intents and purposes, as good as random. All chaotic systems similarly eat up their information and predictability through sensitivity to initial conditions, and after a certain time, the trajectories and numbers that characterize them are unpredicable. In Schopenhauer's worldview, the chance simultaneity of uncountable deterministic causes might underlie

4. The translation of Kierkegaard's choice of the archaic Danish *fuldelig Ro* is open to some controversy. Harder [Har96] offers "sufficient quiet." My Danish friends remind me that the other meaning for this phrase can be "completely drunken."

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events in our world, but the fact of concealed determinism offers no practical advantage over a more stochastic worldview. Schopenhauer's transcendant fatalism has a nice way of supporting Taleb's transcendant fatalism.

Randomness in this book, our *uncertainty*, is *unknowledge* or incomplete information in the sense of Taleb [Tal07]. It is whatever you have not embodied in your state estimation model.

Schopenhauer envisioned each of us decoding and interpreting the web of events around us as "One man's fate is always in keeping with another's, and everyone is the hero of his own drama, but at the same time figures also in that of another ... and that all plurality of phenomena is conditioned by time and space" [SP74]. To the extent that our minds make sense of the external world, and that more than one of our minds agree on such interpretations, depends strongly on our internal models drawing similar state estimations.

Hawking and Mlodinow present the opposite viewpoint, one biased by their inquiry into the nature of physical law. "There may be different ways in which one could model the same physical situation, with each employing different fundamental elements and concepts. If two such physical theories or models accurately predict the same events, one cannot be said to be more real than the other; rather, we are free to use whichever model is most convenient" [HM10]. They will further assume the plausible existence of multiple coexisting realities, a cosmological perspective lacking in the less radical philosophical works to which we now return.

Jung envisioned synchronizing observers. He defines a term *synchronicity*, as "the simultaneous occurrence of a certain psychic state with one or more external events which appear as meaningful parallels to the momentary subjective state—and, in certain cases, vice versa" [JP55a]. In chapter 8 we discussed in detail the equivalence of state estimation with synchrony of the observer and the object observed.

Jung also discusses the Taoist view of the unconscious as a microcosm of macrocosmic events—clearly an allusion to our model-based state estimation. He also discusses the monads of Gottfried Leibnitz's monadology [LL25], wherein each monad is a *little world*: "minds are also images ... capable of knowing the system of the universe, and to some extent of imitating it" [LL25].[5] To Jung, "The perceptive faculty of the monad corresponds to the knowledge, and its appetitive faculty to the will" [JP55a].

Jung stated that "I need hardly point out that for the primitive mind, synchronicity is a self-evident fact" [JP55a]. Unfortunately, today, proving synchronization, especially for complex nonlinear functions, is a question at the forefront of dynamical physics [PCH95, NPS04, NCPS06]. The Taoist concept of meaningful coincidence might, indeed, now be

5. The reader might pause to reflect on how one might attempt to draw analogies from Leibniz's monadology, to the situation of envisioning how individual neurons, as irreducible elements within the brain, interact with each other only through their limited perceptions of their external world (personal communication, J. Y. Lettvin). Such considerations can draw one toward studying neuroscience, as many of us in Professor Lettvin's class were.

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embodied in Hebb's principle of learning—neurons that fire together, wire together—by strengthen their connections [Heb49]. We know today that a precise timing is required for such Hebbian learning to take place. There must be a causality between the cause and the effect [BP98]. Spike timing-dependent plasticity [CD08] is an embodiment of the meaningful coincidence.

Synchrony implies that a functional relationship exists between dynamical systems (chapter 8). Insufficient coupling between systems does not guarantee synchronization. But any degree of coupling does imply that the system receiving information from the other contains information from the signaling system. At sufficient levels of coupling, a system that drives another will become synchronized with it; at that point, *each* system contains enough information to estimate the state of the other [SSC+96]. Coupled dynamical systems live in states of partial information. Synchronized systems understand each other. Our neurons are coupled to each other. The components of our brain are coupled to each other. Our minds are coupled to each other.

Jung discusses the philosophy of Plato, "which takes for granted the existence of transcendental images or models of empirical things, the $\epsilon\iota\delta\eta$ (forms, species), whose reflections ($\epsilon\iota\delta\omega\lambda\alpha$) we see in the phenomenal world" [JP55a]. Plato understood model-based control.

Wolfgang Pauli provided an essay published with Jung's 1955 text [JP55b]. He speculated that

> The process of understanding nature . . . [is] based on a correspondence, a "matching" of inner images pre-existent in the human psyche with external objects and their behavior. This interpretation of scientific knowledge, of course, goes back to Plato and is, as we shall see, very clearly advocated by Kepler. W. Pauli in [JP55b].

Pauli goes on to propose an interpretation of knowledge, as "a 'matching' of external impressions with pre-existent inner images" [JP55b].

Perhaps state estimation of our minds will enable us to better conceive of our minds and society.

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